

Local Overlap Governs Worst-Group Reliability under Covariate Shift:

Why Global Effective Sample Size Misses False Subgroup Alarms

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Abstract

Worst-group reliability is increasingly used to audit models under deployment shift. The usual intuition is simple: if the model fails badly on one subgroup, an average metric can hide that failure. This paper shows that there is a second problem. Under covariate shift, a worst-group audit can also report a subgroup failure that is not real, because the weighted information available for that subgroup may collapse locally even when the global weighted sample still looks healthy.

We study binary prediction under covariate shift and write target subgroup-bin calibration gaps as importance-weighted source quantities. We then identify the right variance scale for these estimators. For each score bin and subgroup, the conditional variance of the self-normalized weighted residual estimator is governed by a subgroup-bin local effective sample size, not by the usual global effective sample size alone. This yields a finite-sample false-alarm bound for worst-group audits in terms of the minimum local effective sample size across the audited cells. We also give a construction showing that global effective sample size can remain large while one subgroup-bin cell has effective sample size one.

Experiments in the repository support the same story. In a pure covariate-shift benchmark where the model is calibrated by construction, lower-tail local effective sample size collapses much faster than global effective sample size, and false worst-group alarms become common. The same pattern persists with learned logistic density-ratio weights. We also retain subgroup-degradation benchmarks showing that real hidden failures do exist in practice. The paper’s message is therefore not to avoid worst-group auditing, but to pair it with local overlap diagnostics. Without that extra check, the audit can confuse statistical noise with deployment failure.

1 Introduction

A model is usually validated on one population and then deployed on another. After deployment, the mix of inputs changes, so the model is no longer being used in the environment where it was checked. This is the familiar distribution-shift problem. In reliability work, the usual concern is that an average calibration score can look fine even when one subgroup is failing badly.

That concern is real, but it is only half of the story.

Consider a hospital risk model that outputs a complication probability for each patient. Suppose one department becomes much more common after deployment. A subgroup-aware audit is a

natural response: instead of looking only at one global calibration number, we inspect departments, workload slices, and their intersections. If one subgroup has a large calibration gap, we would like to treat that as evidence of a real deployment problem.

The difficulty is that this logic mixes two different phenomena:

- a *real hidden failure*, where the model is genuinely unreliable on one subgroup;
- a *false alarm*, where the weighted estimate for one subgroup is unstable because that subgroup has very little effective support after reweighting.

This paper is about separating those two cases.

We work in the standard covariate-shift setting. The source distribution is P , the target distribution is Q , the feature vector is X , the label is $Y \in \{0, 1\}$, and the model outputs a score

$$s(X) \in [0, 1].$$

If the model says 0.8, we would like that to mean “about 80% chance of class 1.” Calibration asks whether predicted probabilities match empirical frequencies.

For one score bin, an intuitive calibration gap is

$$\text{gap} \approx |\text{observed frequency} - \text{predicted probability}|.$$

For example, if the model predicts 0.8 for many patients but only 60% of them actually have the event, then the calibration gap in that part of the score range is about 0.2.

Under covariate shift, target averages can be written as weighted source averages. The basic weight is the density ratio

$$w(x) = \frac{dQ_X}{dP_X}(x),$$

which tells us how much more common x is in deployment than in the source sample. This gives the usual transport rule

$$\text{target average} \approx \frac{1}{n} \sum_{i=1}^n w(X_i) h(X_i, Y_i)$$

for an appropriate quantity h .

At first sight, this suggests a simple solution: estimate worst-group reliability by weighting the source sample and then report the largest subgroup gap. The standard instability diagnostic for this step is the global effective sample size

$$n_{\text{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

If n_{eff} is small, the weighted analysis is noisy. If n_{eff} is large, the audit may look trustworthy.

The main claim of the paper is that this global diagnostic is not enough for worst-group auditing.

Worst-group reliability is built from many subgroup-bin cells. A cell can have almost no weighted support even when the full weighted sample still looks acceptable. The right object is therefore a *local* effective sample size. For one score bin b and one subgroup g , define the active cell indicator

$$A_i(b, g) = \mathbf{1}\{s(X_i) \in I_b\}g(X_i),$$

and the corresponding local effective sample size

$$n_{\text{eff},b,g} = \frac{(\sum_i w_i A_i(b, g))^2}{\sum_i w_i^2 A_i(b, g)}.$$

This formula is exactly the usual effective sample size formula, but restricted to one subgroup-bin cell. It answers a more relevant question:

after weighting, how much information is left for the specific cell that drives the worst-group audit?

This shift from global ESS to local ESS changes the paper’s story.

1. Weighted transport is still the starting point, because it connects source data to target reliability under covariate shift.
2. The main statistical bottleneck is local overlap, because worst-group reliability is determined cell by cell.
3. A subgroup-aware audit can fail in two directions: it can miss a real hidden failure, but it can also create a false subgroup alarm when local weighted support is too weak.

The paper keeps both sides in view. We still prove the usual hidden-failure statement: aggregate calibration can look perfect while subgroup calibration is poor. But the more novel result is about the other direction. We show that for a subgroup-bin residual estimator, the exact conditional variance scale is $1/n_{\text{eff},b,g}$. This leads to a finite-sample false-alarm bound for worst-group audits in terms of the minimum local effective sample size across audited cells. We also construct weighted samples where global ESS remains large while one subgroup-bin cell has local ESS equal to one.

The experiments mirror the same point. In a pure covariate-shift benchmark generated from the repository, the model is calibrated by construction. The true worst-group calibration error is therefore zero. Even in that clean setting, false worst-group alarms become common as shift severity grows, because lower-tail local ESS collapses much faster than global ESS. The same pattern survives with learned logistic density-ratio weights. We then use the existing subgroup-degradation benchmarks in the repository to show that real hidden failures also occur, so the conclusion is not to avoid worst-group auditing. The conclusion is to audit worst-group reliability together with local overlap diagnostics.

1.1 Main Results

Result 1: target subgroup-bin reliability is transported by importance weighting. Under covariate shift, both subgroup-bin residual mass and subgroup-bin conditional calibration gap can be written as weighted source quantities. This is the formal bridge from source data to target reliability.

Result 2: subgroup-bin variance is governed by local effective sample size. For the self-normalized weighted subgroup-bin residual estimator, the conditional variance is controlled by a subgroup-bin local ESS. This is the main technical point in the paper.

Result 3: false worst-group alarms are controlled by the minimum local ESS, not by global ESS alone. When the model is perfectly calibrated on the audited cells, the probability that a worst-group audit exceeds a threshold is bounded by a sum of inverse local ESS terms. A global ESS diagnostic cannot replace this bound.

Result 4: global ESS can be badly misleading. We construct weighted samples where one subgroup-bin cell is effectively supported by a single weighted point while the overall weighted sample still has large global ESS.

Result 5: experiments show both sides of the problem. In pure covariate shift, weak local overlap creates false subgroup alarms. In separate subgroup-degradation benchmarks, genuine hidden failures remain large. The right lesson is therefore not to choose between aggregate and subgroup metrics, but to pair subgroup auditing with local support diagnostics.

2 Problem Setting

2.1 Source data, target data, and covariate shift

Let P be the source distribution and Q the target distribution. The feature vector is $X \in \mathcal{X}$, the label is $Y \in \{0, 1\}$, and the model outputs a fixed score function

$$s : \mathcal{X} \rightarrow [0, 1].$$

We observe labeled source data

$$(X_1, Y_1), \dots, (X_n, Y_n) \sim P$$

and unlabeled or lightly labeled target covariates

$$X'_1, \dots, X'_m \sim Q_X.$$

Assumption 1 (Covariate shift). The conditional label law is unchanged:

$$P(Y | X) = Q(Y | X),$$

while the marginal feature law may change:

$$P_X \neq Q_X.$$

Assumption 2 (Overlap). The target covariate distribution is absolutely continuous with respect to the source covariate distribution, so the density ratio

$$w(x) = \frac{dQ_X}{dP_X}(x)$$

exists and is finite almost surely under P .

These assumptions are standard in covariate-shift adaptation (Sugiyama et al., 2007; Polo and Vicente, 2023). They matter here because they make target reliability quantities transportable through weighted source expectations.

2.2 Groups, bins, and cell-level reliability

Let $\mathcal{G}$ be a finite family of subgroup indicators $g : \mathcal{X} \rightarrow \{0, 1\}$. Let

$$\Pi = \{I_1, \dots, I_B\}$$

be a fixed partition of the score range $[0, 1]$.

For a score bin I_b and subgroup $g \in \mathcal{G}$, define the active cell indicator

$$A_{b,g}(x) = \mathbf{1}\{s(x) \in I_b\}g(x).$$

This is the basic object in the paper: it picks out one score bin and one subgroup at the same time.

Example 1 (Operational subgroups). In a hospital setting, g may represent

$$g_{\text{cardiology}}(x) = \mathbf{1}\{\text{department} = \text{cardiology}\}$$

or an interaction such as

$$g_{\text{urgent,high-load}}(x) = \mathbf{1}\{\text{channel} = \text{urgent}\}\mathbf{1}\{\text{load} \geq c\}.$$

If $I_b = [0.8, 1]$, then $A_{b,g}(x) = 1$ means: this patient is in subgroup g and the model predicted a high score.

We use three related target quantities.

First, the target cell mass is

$$\mu_{b,g}^Q = \mathbb{E}_Q[A_{b,g}(X)]. \quad (2.1)$$

This is simply the target probability of landing in cell (b, g) .

Second, the target residual mass is

$$\theta_{b,g}^Q = \mathbb{E}_Q[(Y - s(X))A_{b,g}(X)]. \quad (2.2)$$

This is the signed calibration error contributed by that cell to the full target population.

Third, when $\mu_{b,g}^Q > 0$, the target cellwise conditional gap is

$$r_{b,g}^Q = \frac{\theta_{b,g}^Q}{\mu_{b,g}^Q} = \mathbb{E}_Q[Y - s(X) \mid A_{b,g}(X) = 1]. \quad (2.3)$$

These three formulas play different roles.

- $\mu_{b,g}^Q$ says how much target mass the cell carries.
- $\theta_{b,g}^Q$ is the cell's contribution to the population-level calibration discrepancy.
- $r_{b,g}^Q$ is the within-cell gap that practitioners would naturally read as a subgroup-bin calibration error.

2.3 Worst-group reliability and hidden failures

We define groupwise target reliability by summing the absolute residual mass across bins:

$$\text{Cal}_{Q,\Pi}(g) = \sum_{b=1}^B |\theta_{b,g}^Q|. \quad (2.4)$$

The aggregate population corresponds to the constant subgroup $\mathbf{1}(x) = 1$. Hence

$$\text{Cal}_{Q,\Pi}(\mathbf{1})$$

is the aggregate binned reliability discrepancy.

The worst-group version is

$$\text{WCE}_{Q,\Pi}(\mathcal{G}) = \max_{g \in \mathcal{G}} \text{Cal}_{Q,\Pi}(g). \quad (2.5)$$

The hidden-failure gap is

$$\text{HFG}_{Q,\Pi}(\mathcal{G}) = \text{WCE}_{Q,\Pi}(\mathcal{G}) - \text{Cal}_{Q,\Pi}(\mathbf{1}). \quad (2.6)$$

These quantities capture the familiar concern that average reliability can hide subgroup failures. The new part of the paper is that the statistical stability of these quantities depends on the local cells (b, g) , not just on the weighted sample globally.

2.4 Weighted estimators and local effective sample size

If oracle weights are available, the natural weighted estimators of [Equations \(2.1\) and \(2.2\)](#) are

$$\hat{\mu}_{b,g} = \frac{1}{n} \sum_{i=1}^n w(X_i) A_{b,g}(X_i), \quad (2.7)$$

and

$$\hat{\theta}_{b,g} = \frac{1}{n} \sum_{i=1}^n w(X_i) (Y_i - s(X_i)) A_{b,g}(X_i). \quad (2.8)$$

Whenever $\hat{\mu}_{b,g} > 0$, the self-normalized cellwise residual estimator is

$$\hat{r}_{b,g} = \frac{\hat{\theta}_{b,g}}{\hat{\mu}_{b,g}} = \frac{\sum_{i=1}^n w(X_i) (Y_i - s(X_i)) A_{b,g}(X_i)}{\sum_{i=1}^n w(X_i) A_{b,g}(X_i)}. \quad (2.9)$$

This is the estimator whose variance is governed by local overlap. It is just a weighted average of residuals inside one subgroup-bin cell.

The corresponding subgroup-bin local effective sample size is

$$n_{\text{eff},b,g} = \frac{(\sum_{i=1}^n w(X_i) A_{b,g}(X_i))^2}{\sum_{i=1}^n w(X_i)^2 A_{b,g}(X_i)}. \quad (2.10)$$

This formula is the usual Kish effective sample size, but applied only to one active cell. It can be read very literally:

- if many weighted points contribute evenly to cell (b, g) , then $n_{\text{eff},g}$ is reasonably large;
- if the cell is effectively supported by one or two large weights, then $n_{\text{eff},g}$ is close to one.

For comparison, the global weighted effective sample size is

$$n_{\text{eff}} = \frac{(\sum_{i=1}^n w(X_i))^2}{\sum_{i=1}^n w(X_i)^2}. \quad (2.11)$$

The paper's main question can now be stated precisely:

which quantity governs the stability of worst-group reliability audits under covariate shift: the global effective sample size n_{eff} , or the local effective sample sizes $n_{\text{eff},g}$?

The next section shows that for subgroup-bin reliability, the relevant variance scale is local.

3 Theory

The theory section answers four questions. First, how are target subgroup-bin quantities transported from the source sample? Second, what controls the variance of a subgroup-bin reliability estimate? Third, how do false worst-group alarms arise under clean covariate shift? Fourth, why is global effective sample size the wrong diagnostic for that problem?

3.1 Transport of subgroup-bin reliability

Proposition 1 (Transport identity for subgroup-bin reliability). *Under [assumptions 1](#) and [2](#), for every bin b and subgroup $g \in \mathcal{G}$,*

$$\mu_{b,g}^Q = \mathbb{E}_P[w(X)A_{b,g}(X)]$$

and

$$\theta_{b,g}^Q = \mathbb{E}_P[w(X)(Y - s(X))A_{b,g}(X)].$$

If $\mu_{b,g}^Q > 0$, then

$$r_{b,g}^Q = \frac{\mathbb{E}_P[w(X)(Y - s(X))A_{b,g}(X)]}{\mathbb{E}_P[w(X)A_{b,g}(X)]}.$$

Remark 1. This is the basic transport step. The numerator is a weighted source average of residuals inside one subgroup-bin cell. The denominator is the corresponding weighted source estimate of the target cell mass. So the cellwise target calibration gap is a ratio of two weighted source expectations. Nothing more complicated is happening yet.

3.2 Local ESS is the correct variance scale

We now study the estimator that matters most for worst-group auditing:

$$\hat{r}_{b,g} = \frac{\sum_{i=1}^n w_i(Y_i - s_i)A_i}{\sum_{i=1}^n w_i A_i},$$

where for short we write $w_i = w(X_i)$, $s_i = s(X_i)$, and $A_i = A_{b,g}(X_i)$.

Assumption 3 (Bounded conditional residual variance). For each audited cell (b, g) , the conditional residual variance

$$\text{Var}(Y - s(X) \mid X = x)$$

is bounded on that cell by a finite constant $\sigma_{b,g}^2$.

Theorem 1 (Local ESS governs subgroup-bin variance). Assume [assumptions 1 to 3](#). Fix a subgroup-bin cell (b, g) and suppose $\sum_{i=1}^n w_i A_i > 0$. Define the conditional weighted mean residual

$$\bar{r}_{b,g}(X_{1:n}) = \frac{\sum_{i=1}^n w_i A_i \mathbb{E}[Y_i - s_i \mid X_i]}{\sum_{i=1}^n w_i A_i}.$$

Then, conditional on $X_1, \dots, X_n$,

$$\mathbb{E}[\hat{r}_{b,g} \mid X_{1:n}] = \bar{r}_{b,g}(X_{1:n})$$

and

$$\text{Var}(\hat{r}_{b,g} \mid X_{1:n}) = \frac{\sum_{i=1}^n w_i^2 A_i \text{Var}(Y_i - s_i \mid X_i)}{(\sum_{i=1}^n w_i A_i)^2} \leq \frac{\sigma_{b,g}^2}{n_{\text{eff},b,g}}. \quad (3.1)$$

The formula is worth reading slowly. Inside one audited cell, $\hat{r}_{b,g}$ is just a weighted average of residuals. The variance of a weighted average is the sum of squared normalized weights times the conditional residual variances. After rearranging, that quantity becomes exactly an inverse local effective sample size. This is why $n_{\text{eff},b,g}$, not the global n_{eff} , is the right information scale.

Remark 2. If the conditional residual variance is nearly constant inside the cell, the upper bound in [Equation \(3.1\)](#) is essentially exact. In that case the local effective sample size is not merely a heuristic diagnostic. It is the variance-determining quantity for the subgroup-bin estimator.

3.3 False worst-group alarms under clean covariate shift

The previous theorem becomes especially sharp when the model is locally well calibrated. That is exactly the setting used in our new synthetic experiment.

Corollary 1 (False-alarm bound for worst-group audits). Assume the conditions of [Theorem 1](#). Suppose that for every audited cell (b, g) ,

$$\mathbb{E}[Y - s(X) \mid X = x] = 0 \quad \text{whenever } A_{b,g}(x) = 1.$$

Then for any threshold $t > 0$,

$$\mathbb{P}\left(\max_{1 \leq b \leq B, g \in \mathcal{G}} |\hat{r}_{b,g}| > t \mid X_{1:n}\right) \leq \sum_{b=1}^B \sum_{g \in \mathcal{G}} \frac{\sigma_{b,g}^2}{t^2 n_{\text{eff},b,g}}. \quad (3.2)$$

In particular, if

$$n_{\text{eff},\min}^{\text{local}} = \min_{1 \leq b \leq B, g \in \mathcal{G}} n_{\text{eff},b,g},$$

then

$$\mathbb{P}\left(\max_{b,g} |\hat{r}_{b,g}| > t \mid X_{1:n}\right) \leq \frac{BG \sigma_{\max}^2}{t^2 n_{\text{eff},\min}^{\text{local}}},$$

where $\sigma_{\max}^2 = \max_{b,g} \sigma_{b,g}^2$.

This is the main statistical message of the paper. Even when there is no true subgroup failure, a worst-group audit can exceed a practical threshold simply because one audited cell has too little weighted information. The audit is then reacting to local overlap collapse, not to a real deployment failure.

3.4 Why global ESS can be non-diagnostic

The next result shows that local and global overlap can tell very different stories.

Proposition 2 (Global ESS can remain large while one audited cell degenerates). *Fix any target cell mass $\pi \in (0, 1)$ and any integer $N \geq 2$. There exists a weighted sample $(A_i, w_i)_{i=1}^N$ such that*

$$\frac{1}{N} \sum_{i=1}^N w_i A_i = \pi, \quad n_{\text{eff cell}} = 1,$$

but the global effective sample size satisfies

$$n_{\text{eff}} = \frac{1}{\pi^2 + (1 - \pi)^2 / (N - 1)}. \quad (3.3)$$

Construction. Take exactly one active point, say $A_1 = 1$, and let $A_2 = \dots = A_N = 0$. Assign weight $w_1 = N\pi$ to the active point and equal weights

$$w_2 = \dots = w_N = \frac{N(1 - \pi)}{N - 1}$$

to the remaining $N - 1$ points. Then the weighted cell mass is π , the active cell contains only one weighted point, and therefore $n_{\text{eff cell}} = 1$. The global ESS is obtained by direct substitution into Equation (2.11), which yields Equation (3.3). $\square$

Remark 3. This proposition is intentionally simple. It says that a subgroup-bin cell can be effectively supported by one weighted point while the full weighted sample still has large global ESS. For example, if $\pi = 0.05$, then the right-hand side of Equation (3.3) is close to 400 for large N . So a reassuring global ESS number does not prevent one audited cell from being statistically degenerate.

3.5 Real hidden failures still matter

The local-ESS story does not replace the hidden-failure story. It complements it.

Proposition 3 (Aggregate calibration can hide genuine subgroup failure). *There exist distributions P, Q , a score function s , and a subgroup family $\mathcal{G}$ such that*

$$\text{Cal}_{Q, \Pi}(\mathbf{1}) = 0 \quad \text{but} \quad \text{WCE}_{Q, \Pi}(\mathcal{G}) > 0.$$

Construction. Take two groups g_A and g_B with equal target mass. Let $Y = 0$ almost surely on group A and $Y = 1$ almost surely on group B . Let the score be constant, $s(X) = 1/2$, and use a single score bin. Then the aggregate residual is

$$\mathbb{E}_Q[Y - s(X)] = \frac{1}{2}(0 - 1/2) + \frac{1}{2}(1 - 1/2) = 0,$$

so aggregate calibration is perfect. But inside each subgroup the absolute calibration gap is $1/2$, so subgroup reliability is poor. $\square$

Remark 4. Proposition 3 explains why worst-group auditing is needed in the first place. Corollary 1 explains why that audit must also be accompanied by local support diagnostics. Together they describe the two-sided statistical problem: the audit can miss a real hidden failure, and it can also create a false one.

3.6 Estimated weights and selective deployment

The local-ESS theorem isolates the sampling-noise part of the problem under oracle weights. With estimated weights $\hat{w}$, the total error decomposes into two pieces:

$$|\hat{r}_{b,g}(\hat{w}) - r_{b,g}^Q| \leq |\hat{r}_{b,g}(\hat{w}) - \hat{r}_{b,g}(w)| + |\hat{r}_{b,g}(w) - r_{b,g}^Q|.$$

The first term is weight-estimation error. The second is the local-overlap term analyzed above. So local ESS does not replace density-ratio estimation diagnostics. It tells us what happens once the weights are fixed.

The same logic also applies to selective deployment. If $\mathcal{A}_\tau = \{x : u(x) \geq \tau\}$ is the accepted set at threshold τ , then the accepted-set subgroup indicator is just

$$\tilde{A}_{g,\tau}(x) = \mathbf{1}\{x \in \mathcal{A}_\tau\}g(x).$$

Hence accepted-set risk, accepted-set calibration, and subgroup abstention gaps are governed by the same local-overlap mechanism. The selective-prediction story is therefore an extension of the same variance geometry, not a separate problem (Geifman and El-Yaniv, 2017; Jones et al., 2020).

4 Experiments

The experiments are organized around the paper’s two-sided claim.

- First, even under pure covariate shift, worst-group audits can create false subgroup alarms when local weighted support collapses.
- Second, real hidden subgroup failures still exist, so the right response is not to stop auditing subgroups but to audit them with local overlap diagnostics.

4.1 Experiment 1: false alarms under pure covariate shift

The main experimental study is new and is saved in `preprint/research/local_ess_study.py`. It uses `src/shiftstat/datasets/synthetic.py` and intentionally chooses the cleanest possible setting: the data are generated under pure covariate shift, and the prediction score is the true conditional probability. In other words,

$$Y | X \sim \text{Bernoulli}(s(X))$$

under both source and target. The model is therefore calibrated by construction.

This design matters because it removes the usual ambiguity. If a worst-group audit reports a large subgroup-bin calibration gap in this experiment, that is not a real deployment failure. It is a false alarm caused by weighted finite-sample noise.

The audit family is intentionally richer than a single pre-specified slice. We use:

- threshold groups based on x_0, x_1, x_2 ;
- negative-tail and positive-tail slices;

Table 1: Main local-ESS experiment. The model is calibrated by construction, so the true worst-group error is zero. Reported worst-group values are therefore false alarms created by weighted noise.

Weights	Shift	Median global ESS	Median lower-tail local ESS	Median false worst-group error	False-alarm rate > 0.05	False-alarm rate > 0.10
Oracle	0.4	312.0	2.76	0.040	0.19	0.00
Oracle	0.8	94.7	1.44	0.070	0.80	0.14
Oracle	1.2	27.2	1.05	0.110	0.89	0.55
Oracle	1.6	10.9	1.00	0.138	0.89	0.69
Logistic	0.8	103.0	1.38	0.071	0.82	0.06
Logistic	1.2	50.2	1.00	0.099	0.92	0.48

- pairwise positive-tail intersections;
- seven score bins.

We run the study in two modes.

- **Oracle weights:** exact density-ratio weights from the Gaussian covariate-shift generator, over 200 random seeds.
- **Learned weights:** logistic density-ratio estimation through the library’s `ImportanceWeighter`, over 50 random seeds at the middle shift severities.

Results at a glance. The main empirical result is simple:

as shift grows, lower-tail local ESS collapses much faster than global ESS, and false worst-group alarms become common even though the true worst-group error is zero.

Local ESS predicts subgroup-bin noise more directly than global ESS. The saved study also reports correlation diagnostics. Under oracle weights, the pooled correlation between $\log(1 + n_{\text{eff},g})$ and the absolute subgroup-bin normalized gap is about -0.49 , whereas the corresponding pooled correlation for global ESS is only about -0.30 . More importantly, within each fixed shift level, lower-tail local ESS predicts worst-group noise more strongly than global ESS. This is the empirical analogue of [Corollary 1](#).

Why the experiment matters. This is not just another covariate-shift benchmark. It isolates the exact failure mode of interest. Since there is no true subgroup miscalibration in the data-generating process, the only reason for a large worst-group estimate is statistical instability. The experiment therefore turns the theorem into something observable: local overlap collapse is not an abstract concern. It changes the behavior of the audit in a measurable way.

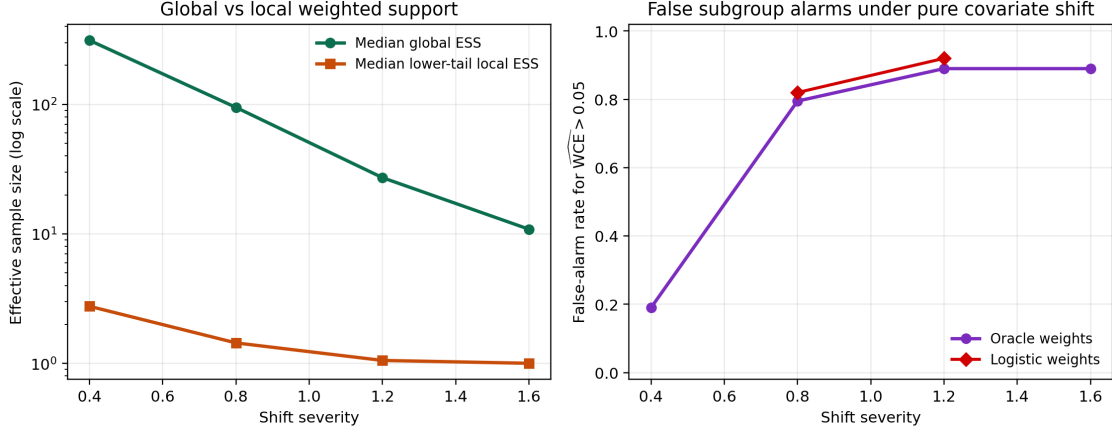


Figure 1: Left: global ESS declines with shift severity, but the lower tail of local ESS collapses much faster. Right: false worst-group alarms become common as shift grows, even though the model is calibrated by construction. Logistic density-ratio weights show the same qualitative pattern.

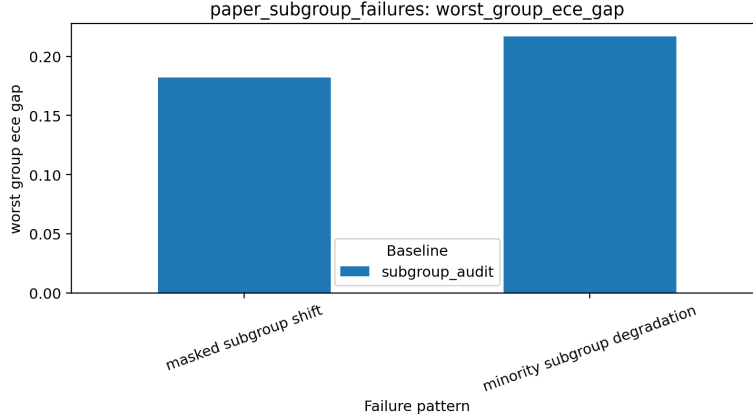


Figure 2: Worst-group ECE gap in the repository’s subgroup-degradation benchmark suite. These are genuine hidden failures, not false alarms.

4.2 Experiment 2: real hidden failures still occur

The local-ESS study explains false alarms. It does not remove the need for subgroup auditing. To show that, we keep the repository’s subgroup-degradation benchmarks as a second experiment.

These scenarios come from `src/shiftstat/datasets/audit_synthetic.py` and are already used by the repository’s publication suite. Here the target process is changed in a subgroup-specific way, so the subgroup failures are real rather than noise-driven. Two representative cases are:

- **Masked subgroup shift:** worst-group ECE gap ≈ 0.182 .
- **Minority subgroup degradation:** worst-group ECE gap ≈ 0.217 .

Interpretation. Taken together, the two experiments say exactly what the theory says.

- Worst-group auditing is necessary because aggregate metrics can hide real subgroup failures.
- Worst-group auditing is not self-authenticating because local overlap collapse can also create false subgroup alarms.

The practical implication is therefore simple: whenever a worst-group audit is reported under covariate shift, the report should also include local support diagnostics such as subgroup-bin effective sample sizes or lower-tail summaries of those quantities.

4.3 Reproducibility

The new local-ESS experiment writes its outputs to:

- `preprint/research/results/local_ess_oracle/`
- `preprint/research/results/local_ess_logistic/`

The supporting hidden-failure figures come from the committed publication artifacts already present in the repository:

- `paper_assets/generated/publication_suite/paper_subgroup_failures/`

5 Discussion

The paper’s story is now sharper than the original draft. Under deployment shift, worst-group auditing is necessary because aggregate calibration can hide real subgroup failures. But under covariate shift, worst-group auditing is also statistically fragile because the relevant information is local. A subgroup-bin cell can become effectively unsupported long before the weighted sample looks globally unusable.

This perspective clarifies three points that are often conflated.

- **Transport is not the same as stability.** Covariate shift tells us how to rewrite target quantities as weighted source quantities. It does not tell us whether those quantities can be estimated reliably in every audited cell.
- **Global ESS is not a worst-group diagnostic.** It summarizes the full weighted sample. Worst-group auditing depends on the weakest audited cell, so the relevant object is the lower tail of local ESS.
- **A subgroup alarm is not automatically evidence of a real deployment failure.** It may reflect true subgroup harm, or it may reflect local overlap collapse. The audit result is only interpretable once both possibilities are on the table.

This leads to a practical recommendation that is easy to state:

report worst-group reliability together with local overlap diagnostics.

In concrete terms, a subgroup audit under covariate shift should not stop at one global ESS value. It should also report cell-level or lower-tail local ESS summaries, ideally for the exact subgroup-bin family used by the audit. Otherwise a reviewer or practitioner cannot tell whether a large worst-group estimate reflects deployment harm or statistical degeneracy.

There are several directions that could make the paper even stronger.

1. Replace the finite subgroup family by a richer class and derive complexity-sensitive local-overlap bounds.
2. Move from point estimates to interval estimates, for example confidence intervals for subgroup-bin gaps or worst-group error conditioned on local ESS.
3. Study adaptive auditing, where the subgroup family itself is chosen after looking at the data, which should make false-alarm control even harder.
4. Extend the local-overlap analysis beyond pure covariate shift to mixed shift, concept drift, and settings with doubly robust correction (Yang et al., 2024; Wu et al., 2024).

The broader claim is not about one library or one benchmark suite. It is a statistical claim about deployment auditing. If worst-group reliability is going to be used as evidence in shifted environments, then local overlap has to be part of the evidence. That is the main message of the paper.

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A Appendix

A.1 Proof of Proposition 1

We prove the two transported moments and then divide them to obtain the ratio form for $r_{b,g}^Q$. For the cell mass, apply the standard change-of-measure identity with

$$h(x, y) = A_{b,g}(x).$$

Then

$$\mathbb{E}_P[w(X)A_{b,g}(X)] = \mathbb{E}_Q[A_{b,g}(X)] = \mu_{b,g}^Q.$$

For the residual mass, choose

$$h(x, y) = (y - s(x))A_{b,g}(x).$$

The same identity gives

$$\mathbb{E}_P[w(X)(Y - s(X))A_{b,g}(X)] = \mathbb{E}_Q[(Y - s(X))A_{b,g}(X)] = \theta_{b,g}^Q.$$

If $\mu_{b,g}^Q > 0$, then Equation (2.3) implies

$$r_{b,g}^Q = \frac{\theta_{b,g}^Q}{\mu_{b,g}^Q} = \frac{\mathbb{E}_P[w(X)(Y - s(X))A_{b,g}(X)]}{\mathbb{E}_P[w(X)A_{b,g}(X)]}.$$

A.2 Proof of Theorem 1

Fix a subgroup-bin cell (b, g) and abbreviate $A_i = A_{b,g}(X_i)$, $w_i = w(X_i)$, and $s_i = s(X_i)$. Conditional on $X_{1:n}$, the denominator

$$S_A = \sum_{i=1}^n w_i A_i$$

is deterministic and positive by assumption. Therefore

$$\hat{r}_{b,g} = \sum_{i=1}^n \alpha_i (Y_i - s_i), \quad \alpha_i = \frac{w_i A_i}{S_A}.$$

So $\hat{r}_{b,g}$ is a weighted average with normalized weights α_i .

Taking conditional expectation gives

$$\mathbb{E}[\hat{r}_{b,g} | X_{1:n}] = \sum_{i=1}^n \alpha_i \mathbb{E}[Y_i - s_i | X_i] = \frac{\sum_{i=1}^n w_i A_i \mathbb{E}[Y_i - s_i | X_i]}{\sum_{i=1}^n w_i A_i} = \bar{r}_{b,g}(X_{1:n}).$$

Because the observations are conditionally independent given the features,

$$\text{Var}(\hat{r}_{b,g} | X_{1:n}) = \sum_{i=1}^n \alpha_i^2 \text{Var}(Y_i - s_i | X_i).$$

Substituting the definition of α_i yields

$$\text{Var}(\hat{r}_{b,g} | X_{1:n}) = \frac{\sum_{i=1}^n w_i^2 A_i^2 \text{Var}(Y_i - s_i | X_i)}{(\sum_{i=1}^n w_i A_i)^2}.$$

Since $A_i \in \{0, 1\}$, we have $A_i^2 = A_i$, so

$$\text{Var}(\hat{r}_{b,g} | X_{1:n}) = \frac{\sum_{i=1}^n w_i^2 A_i \text{Var}(Y_i - s_i | X_i)}{(\sum_{i=1}^n w_i A_i)^2}.$$

Using the bound $\text{Var}(Y - s(X) | X = x) \leq \sigma_{b,g}^2$ on the active cell gives

$$\text{Var}(\hat{r}_{b,g} | X_{1:n}) \leq \sigma_{b,g}^2 \frac{\sum_{i=1}^n w_i^2 A_i}{(\sum_{i=1}^n w_i A_i)^2} = \frac{\sigma_{b,g}^2}{n_{\text{eff},b,g}}.$$

This proves [Theorem 1](#).

A.3 Proof of [Corollary 1](#)

Under the additional assumption in [Corollary 1](#), we have

$$\mathbb{E}[Y_i - s_i | X_i] = 0$$

whenever $A_i = 1$. By [Theorem 1](#),

$$\mathbb{E}[\hat{r}_{b,g} | X_{1:n}] = 0$$

for every audited cell with positive denominator. Chebyshev's inequality therefore gives

$$\mathbb{P}(|\hat{r}_{b,g}| > t | X_{1:n}) \leq \frac{\text{Var}(\hat{r}_{b,g} | X_{1:n})}{t^2} \leq \frac{\sigma_{b,g}^2}{t^2 n_{\text{eff},b,g}}.$$

Applying a union bound over all $B|\mathcal{G}|$ audited cells yields [Equation \(3.2\)](#). The second display in the corollary follows by replacing each local ESS by the smallest one and each variance bound by $\sigma_{\max}^2$.

A.4 Further explanation of [Proposition 2](#)

The counterexample is deterministic because the point does not depend on any stochastic subtlety. The sample can be globally well weighted and locally degenerate at the same time.

Take one active cell point with weight $N\pi$ and $N-1$ inactive points with equal weight $N(1-\pi)/(N-1)$. Then:

- the weighted target mass of the active cell is exactly π ;
- the active cell is supported by only one weighted point, so its local ESS is exactly 1;
- the full sample still spreads most of its mass across many inactive points, so the global ESS can remain large.

This construction is enough to show that a large global ESS does not rule out a statistically meaningless subgroup-bin estimate.

A.5 Further explanation of [Proposition 3](#)

The hidden-failure construction and the local-ESS construction diagnose different problems.

- [Proposition 3](#) is about *real hidden harm*: opposite-sign subgroup errors cancel in the aggregate.
- [Proposition 2](#) and [Corollary 1](#) are about *false subgroup alarms*: the subgroup estimate is unstable because the cell has too little weighted information.

The final paper needs both statements. Without [Proposition 3](#), there is no reason to audit subgroups. Without [Corollary 1](#), there is no reason to distrust an unstable subgroup alarm.

A.6 Experimental provenance

The new local-ESS study is generated by:

- `preprint/research/local_ess_study.py`

The rendered outputs used in the main text are:

- `preprint/research/results/local_ess_oracle/`
- `preprint/research/results/local_ess_logistic/`
- `preprint/figures/local_ess_main.png`

The supporting hidden-failure figure comes from:

- `paper_assets/generated/publication_suite/paper_subgroup_failures/`